

PARTH GOYAL

Quantitative Researcher · Economics Writer

✉ parthgoyal60@gmail.com 🌐 parthgoyal.uk 🔗 linkedin.com/in/parthg1206 🐙 github.com/ParthG60 📖 parthstack.substack.com

Education

University of St Andrews

Sep 2020 – Jun 2024

BSc (Honours) Financial Economics and Mathematics, Upper Second Class (2:1)

St Andrews, UK

- **Dissertation (Mathematics):** *Modeling the Divide* — built a 5,000-agent heterogeneous ABM (partisan/elite types, ANES-initialised opinions, Gini-scaled tolerance) showing exposure and initial-distribution asymmetry drive bimodal polarization; independents and lower inequality dampen divergence [live app]
- **Relevant Modules:** Statistical Inference, Econometrics, Game Theory, Asset Pricing, Quantitative Risk Management, Time Series Analysis, Bayesian Statistics, Corporate Finance and Portfolio Management, Micro & Macroeconomics, Probability and Combinatorics, Linear Algebra, Multivariate Calculus, OO Programming, Differential Equations

CFA Institute

CFA Level I — Passed (Top 5% globally)

Experience

Orbis Investments

Sep 2024 – Present

Quantitative Researcher

London, UK

- Designed and own systematic stock-selection signals for global financials, using fundamental line-items to predict forward returns; backtests at ~10% annual outperformance and is used in production for market-neutral portfolio construction (~\$1bn AUM)
- Built a novel time-series factor model for a concentrated US equity portfolio that explains ~80% of return variance (adj. R^2); used by the portfolio manager to diagnose individual bets and size hedging trades
- Use LLM APIs (Claude, GPT) to extract qualitative insights at scale from earnings-call transcripts and analyst reviews — building a structured signal layer over text that the team couldn't easily reach manually
- *Prior summer internship (2023):* researched alternative stock-beta estimators (Bayesian, robust, time-decayed) over 40 years of US equity data, improving on the firm's prior standard (~1% on volatility); built an interpretable XGBoost model of portfolio-manager behavioural biases to flag likely losing trades (walk-forward validation, SHAP)

St Andrews AI Safety (STAISH)

Spring 2024

Co-founder & Program Director

St Andrews, UK

- Co-founded the University of St Andrews' first AI safety fellowship — a 6-week discussion program covering ML fundamentals, AGI risk, reward misspecification and goal misgeneralisation, AI & the economy, and governance
- Curated and synthesised 30+ readings and talks across the alignment, technical, and policy literature (Ngo, Hubinger, Yudkowsky, DeepMind, OpenAI, Dafoe); designed and facilitated weekly discussion questions
- Cohort outcomes include one participant taking an OpenAI internship, another moving into AI safety research, and several directing their undergraduate dissertations to AI-related topics

St Andrews Effective Altruism Society

Sep 2022 – Jun 2024

Treasurer and Tutor

St Andrews, UK

- Successfully secured a £12,000 grant for the society, with a focus on optimizing fund allocation for maximum impact
- Led discussions on crucial Effective Altruism (EA) concepts such as decision-making, risk assessment, and longtermism
- Conducted weekly workshops, imparting key technical skills such as forecasting, argument framing, and charity effectiveness analysis, resulting in a 50% increase in society enrollment

Skills

Technical: Python [Pandas, NumPy, SciPy, scikit-learn, XGBoost, SHAP, cvxpy, LLM APIs], R [tidyverse, ggplot2, fixest, time-series], SQL (Snowflake), LaTeX

Languages: Russian, English, Hindi